

Unifying Theory of Dynamic Aether Mechanics: Bound Core Electron Model

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Working Paper in the Dynamic Aether Mechanics Framework

Abstract

Paper 7 [6] identifies the electron as a quantized vortex ring in the Aether condensate, with mass determined by the Donnelly kinetic energy formula. The vortex core is treated as a depleted region where the condensate density drops to zero at the healing length ξ . This paper proposes a refinement: the Dynamic Casimir Effect (DCE), which operates at the vortex boundary, creates bound Aether that migrates inward under the Bernoulli pressure gradient and accumulates as a toroidal ring in the vortex core. The bound ring grows until the boundary velocity drops below the DCE permanence threshold, at which point only transient binding (gravity) occurs and the electron mass stabilizes. This self-regulating mechanism explains why the electron has the mass it does. The self-consistent vortex equation, corrected for core energy (+5–14%) and the Bernoulli density deficit (–1.7%), predicts $m_e \approx 0.4\text{--}0.6$ MeV across the allowed parameter range and matches the observed 0.511 MeV for $m_A \approx 628\text{--}689$ MeV/ c^2 , well within the independently constrained convergence zone.

This working paper explores a specific mechanism within the Dynamic Aether Mechanics framework. It references results from Paper 1 (General Overview) [1], Paper 2 (Special Relativity) [2], Paper 3 (Gravity) [3], Paper 4 (Magnetic and Electric Fields) [4], Paper 5 (Strong and Weak Forces) [5], and Paper 7 (Quantum Mechanics) [6] of the main series. The ideas presented here are under development and may be incorporated into the main papers in revised form.

1. Introduction

Paper 7 [6] establishes that the electron is a quantized vortex ring in the Aether condensate with circulation $\kappa_e = h/(2m_A)$. The electron rest energy is set equal to the Donnelly kinetic energy of the circulatory flow:

$$m_e c^2 = \frac{1}{2} \rho_A \kappa_e^2 r_{\text{loop}} \left[\ln \left(\frac{8 r_{\text{loop}}}{\xi} \right) - \frac{1}{2} \right] \quad (1)$$

where $r_{\text{loop}} = \hbar/(m_e c)$ is the loop radius (reduced Compton wavelength), $\xi = \hbar/(m_A c \sqrt{2})$ is the healing length, and the core is treated as a depleted region where the condensate density drops to zero.

This treatment leaves two questions open:

1. **Where does the DCE product go?** Paper 1 [1] establishes that the spinning vortex core drives the Dynamic Casimir Effect: transient binding produces gravity, and permanent binding (above the DCE threshold) produces mass. But the permanently bound material must go somewhere. The current model is silent on its fate.
2. **Why does the electron have the mass it does?** Eq. (1) determines m_e given m_A and ρ_A , but provides no mechanism for *why* this particular vortex energy is selected from among all possible configurations.

This paper proposes that both questions have the same answer: the DCE permanently binds Aether at the vortex boundary, the bound material migrates inward under the Bernoulli pressure gradient and accumulates as a toroidal ring in the core, the ring grows until the boundary velocity drops below the permanence threshold, and the electron mass stabilizes at this self-regulating equilibrium.

2. Formation Model

The electron forms through the following sequence:

1. **Vortex ring creation.** In the high-energy early universe, turbulent Aether flows create quantized vortex rings with the minimum (spin- $\frac{1}{2}$) circulation quantum $\kappa_e = h/(2m_A)$. Initially these are pure fluid vortices with no bound material—the core is depleted at the healing length as in the standard model.
2. **DCE creates matter.** The circulation velocity at the healing length is (Section 3.1):

$$v(\xi) = \frac{c}{\sqrt{2}} \approx 0.707 c \quad (2)$$

This exceeds the DCE permanence threshold, so the rapidly-circulating boundary continuously converts vacuum fluctuations into permanent bound Aether [8].

3. **Matter migrates inward.** The Bernoulli pressure in the vortex is lowest at the core (Section 3.2). Bound material, no longer part of the superfluid flow, is pushed inward by the pressure gradient and accumulates along the vortex tube axis, forming a toroidal ring of bound Aether tracing the major radius r_{loop} .
4. **Ring grows.** As more bound material falls to the center, the ring expands radially outward from the tube axis. The effective core boundary moves from ξ to a larger radius r_{eq} .

5. **Boundary velocity drops.** The circulation κ_e is topologically quantized and fixed, but the velocity at the boundary is $v(r) = \kappa_e / (2\pi r)$. As r_{eq} increases, the boundary velocity decreases.
6. **Permanence threshold reached.** When $v(r_{\text{eq}})$ drops to v_{perm} —the minimum velocity for permanent DCE binding—the vortex can no longer create permanent mass. Only transient bind–unbind cycling occurs at the boundary, producing the asymmetric pressure oscillation that manifests as gravity [3].
7. **Mass stabilizes.** The electron’s mass is the sum of the remaining circulation kinetic energy and the bound ring’s rest mass at this equilibrium point.

This is a *self-regulating* process: the creation of mass enlarges the core, which reduces the boundary velocity, which eventually shuts off mass creation. The electron mass is not a free parameter but an equilibrium outcome determined by medium properties.

2.1 Energy Conservation

The DCE mechanism converts energy from the vortex’s circulation into bound material—the moving boundary does work on the vacuum fluctuations, and the energy is drawn from the kinetic energy of the flow. As the ring grows, the circulation KE decreases by the same amount that the ring mass increases. The total energy is approximately conserved during the formation process:

$$m_e c^2 \approx E_{\text{KE,initial}} = E_{\text{KE,final}} + m_{\text{ring}} c^2 \quad (3)$$

The electron’s final mass is set by the initial vortex energy at creation, not by the partitioning between KE and ring mass. The circulation quantum $\kappa_e = h / (2m_A)$ is topologically fixed and cannot change; the KE reduction comes from the shrinking integration domain as the core expands from ξ to r_{eq} .

3. Key Equations

3.1 Circulation Velocity

For a quantized vortex with circulation $\kappa_e = h / (2m_A) = \pi \hbar / m_A$, the flow velocity at distance r from the tube axis is:

$$v(r) = \frac{\kappa_e}{2\pi r} \quad (4)$$

At the healing length $\xi = \hbar / (m_A c \sqrt{2})$, the velocity evaluates to $v(\xi) = c / \sqrt{2} \approx 0.707 c$ (Eq. 2). This result is exact and independent of m_A : the circulation quantum and the healing length are both proportional to $1/m_A$, so m_A cancels.

3.2 Bernoulli Pressure

The pressure distribution in the vortex flow:

$$P(r) = P_\infty - \frac{1}{2} \rho_A v^2(r) = P_\infty - \frac{\rho_A \kappa_e^2}{8\pi^2 r^2} \quad (5)$$

Pressure is lowest at the core center and increases outward. Bound material (not participating in superfluid flow) experiences a net inward force from the surrounding higher-pressure fluid, driving inward migration.

3.3 Equilibrium Condition

The bound ring grows until the boundary velocity reaches the permanence threshold:

$$v(r_{\text{eq}}) = v_{\text{perm}} \implies r_{\text{eq}} = \frac{\kappa_e}{2\pi v_{\text{perm}}} \quad (6)$$

For $v_{\text{perm}} < c/\sqrt{2}$, the ring extends beyond the healing length: $r_{\text{eq}} > \xi$.

4. The Electron Mass Equation

4.1 Core Energy

The Donnelly formula [7] (Eq. 1) computes only the kinetic energy of the circulating flow outside the core. In a standard GPE vortex, the total energy also includes the core's gradient and potential energy—the cost of depleting the condensate. This core energy is of order $1/\ln(r_{\text{loop}}/\xi)$ of the kinetic energy, or $\sim 5\text{--}14\%$ for the electron's aspect ratio (the exact value for the logarithmic potential used in this theory has not been computed from first principles).

In the bound-core model, the core region is filled with bound material instead of depleted vacuum. The bound ring's rest mass *replaces* the gradient and potential energy of the depleted core—it is the physical realization of the same core energy, not an additional contribution on top of it.

4.2 Bernoulli Density Deficit

The Donnelly formula assumes constant density ρ_A everywhere. But the vortex circulation itself produces a Bernoulli density deficit. This is a genuine density change (not just a pressure change with constant ρ_A) because the vortex circulation is *driven* flow, not free-fall—the persistent spin energy maintains the flow against Bernoulli pressure, creating a venturi-like deficit. This is physically

distinct from gravitational inflow, where free-fall dynamics ensure $\Delta\rho = 0$ exactly [2]:

$$\frac{\delta\rho(r)}{\rho_A} \approx -\frac{v^2(r)}{2c^2} = -\frac{\kappa_e^2}{8\pi^2 c^2 r^2} \quad (7)$$

At the healing length ($v = c/\sqrt{2}$), the deficit reaches $\delta\rho/\rho_A \approx -25\%$, dropping rapidly with distance:

r/ξ	v/c	$\delta\rho/\rho_A$
1.0	0.707	-25%
1.5	0.471	-11%
2.0	0.354	-6%
5.0	0.141	-1%

The density deficit reduces the kinetic energy. Integrating the correction $\delta E = \int \frac{1}{2} (-\delta\rho) v^2 dV$:

$$\frac{\delta E_{\text{KE}}}{E_{\text{KE}}} \approx -\frac{1}{8 \ln(r_{\text{loop}}/\xi)} \approx -1.7\% \quad (8)$$

The factor $1/8$ arises because $\kappa_e^2/(16\pi^2 c^2 \xi^2) = 1/8$ for $v(\xi) = c/\sqrt{2}$. For the logarithmic equation of state, the exact compressible Bernoulli gives $\rho(r) = \rho_A e^{-v^2/(2c^2)}$ rather than the linearized $\rho_A [1 - v^2/(2c^2)]$; the integrated KE correction differs by less than 3% between exact and linear forms, so the linearized result is adequate.

The Donnelly formula itself is non-relativistic. In the Aether framework, relativistic effects emerge from the medium at $O(v^2/c^2)$: time dilation from wave path geometry, length contraction from density changes [2]. The Bernoulli density correction is the leading such correction to the vortex energy; higher-order terms are $O(v^4/c^4) \sim 0.4\%$, negligible at this level of precision.

4.3 Combined Mass Equation

Combining all contributions:

$$m_e c^2 \approx \underbrace{\frac{1}{2} \rho_A \kappa_e^2 r_{\text{loop}} \left[\ln\left(\frac{8 r_{\text{loop}}}{r_{\text{eq}}}\right) - \frac{1}{2} \right]}_{E_{\text{KE}} \text{ (circulation)}} \times \underbrace{(1 - 0.017)}_{\text{Bernoulli}} + \underbrace{E_{\text{core}}}_{\sim 5-14\% \text{ of } E_{\text{KE}}} \quad (9)$$

The net correction from the bare Donnelly formula is:

$$m_e c^2 \approx E_{\text{KE}}^{\text{Donnelly}} \times (1 - 0.017 + f_{\text{core}}) \quad (10)$$

where $f_{\text{core}} \approx 0.05\text{--}0.14$ is the core energy fraction (model-dependent). The net correction ranges from +3% to +12% above the bare Donnelly value. The loop radius is $r_{\text{loop}} = \hbar/(m_e c)$ (the reduced Compton wavelength), following Paper 7 [6].

5. Mass Predictions

5.1 Donnelly Kinetic Energy

Using the self-consistent condition $r_{\text{loop}} = \hbar/(m_e c)$ and setting $r_{\text{eq}} = \xi$, the Donnelly formula gives:

$$m_e^2 = \frac{\pi^2 \rho_A \hbar^3}{2 m_A^2 c^3} \left[\ln \left(\frac{8\sqrt{2} m_A}{m_e} \right) - \frac{1}{2} \right] \quad (11)$$

This transcendental equation is solved iteratively. Results for $\rho_A = 5 \times 10^{11} \text{ kg/m}^3$:

m_A (MeV/ c^2)	ξ (fm)	r_{loop} (fm)	m_e (KE only) (MeV)	vs. observed
500	0.28	325	0.61	+19%
550	0.25	354	0.56	+9%
600	0.23	383	0.515	+1%
650	0.21	411	0.48	−6%
700	0.20	439	0.45	−12%
800	0.17	495	0.40	−22%

The bare Donnelly formula gives $m_e \approx 0.4\text{--}0.6 \text{ MeV}$ across the full convergence zone (500–850 MeV/ c^2 from Paper 7 [6]), bracketing the observed 0.511 MeV.

5.2 Corrected Prediction

Since DCE converts circulation KE into ring mass (energy conserved, Eq. 3), the total electron mass is set by the initial vortex energy. The Donnelly formula gives only the KE portion. Including the core energy and Bernoulli corrections (Eq. 10), the corrected prediction depends on the core energy fraction f_{core} :

f_{core}	Net correction	Best-fit m_A	KE only (MeV)	Corrected (MeV)
5%	+3%	628 MeV	0.495	0.511
8%	+6%	649 MeV	0.480	0.510
11%	+9%	669 MeV	0.468	0.512
14%	+12%	689 MeV	0.455	0.511

For any core energy in the 5–14% range, there exists an m_A in the convergence zone that matches the observed mass:

$$m_e \approx 0.511 \text{ MeV} \quad (\text{predicted, } m_A \approx 628\text{--}689 \text{ MeV}) \quad \text{vs.} \quad m_e = 0.511 \text{ MeV} \quad (\text{observed}) \quad (12)$$

The Bernoulli density deficit (−1.7%) partially offsets the core energy (+5–14%), and the remaining net correction (+3–12%) is absorbed by m_A , which shifts from 600 MeV (bare Donnelly) to 628–689 MeV (corrected).

6. Physical Structure at Equilibrium

6.1 Compression Ratio

Even if the entire electron mass were packed into the core ring (the extreme $f = 1$ case), the compression ratio would be remarkably low:

$$X = \frac{\rho_{\text{core}}}{\rho_A} = \frac{m_e}{\rho_A \cdot 2\pi^2 r_{\text{loop}} \xi^2} \approx 4\text{--}5 \quad (13)$$

(varying from ~ 4.4 at $m_A = 600$ to ~ 4.9 at $m_A = 630$). Only ~ 5 times ambient density even in the extreme case—far below the $\sim 7 \times 10^7$ compression in QCD bridges [5]. This reflects the electron’s small mass (0.511 MeV) spread over a large circumference ($2\pi r_{\text{loop}} \approx 2400$ fm).

For the core energy range $f_{\text{core}} = 5\text{--}14\%$, the ring carries only 0.03–0.07 MeV. Spread over the core volume ($\sim 300\text{--}400 \text{ fm}^3$), the ring density is well below ρ_A —the core is sparsely filled, not compressed.

6.2 Self-Regulation

The self-regulation mechanism proceeds as follows:

1. DCE creates bound material at the core boundary where $v > v_{\text{perm}}$
2. Material falls inward, expanding the effective core from ξ to r_{eq}

3. At r_{eq} , the velocity has dropped: $v(r_{\text{eq}}) < v(\xi)$
4. When $v(r_{\text{eq}}) = v_{\text{perm}}$, permanent binding ceases
5. Transient binding continues (producing gravity), but no new mass accumulates

The quantized circulation $\kappa_e = h/(2m_A)$ is topologically fixed—it cannot change. Only the boundary radius changes as the ring grows, reducing the velocity through the $v \propto 1/r$ dependence.

Under energy conservation (Eq. 3), the total electron mass is fixed by the initial vortex energy at creation. The threshold velocity v_{perm} determines only the partitioning between circulation KE and ring rest mass, not the total. The ring mass at equilibrium is:

$$m_{\text{ring}} c^2 = f_{\text{core}} \times m_e c^2 \quad (14)$$

where $f_{\text{core}} \approx 5\text{--}14\%$. The electron remains circulation-dominated at equilibrium.

6.3 The DCE Permanence Threshold

The permanence threshold v_{perm} determines the equilibrium core size $r_{\text{eq}} = \kappa_e/(2\pi v_{\text{perm}})$, the KE/ring mass partitioning, and the detailed structure of the electron. For v_{perm} near $c/\sqrt{2}$ (modest ring growth), the total mass varies only weakly with v_{perm} . A natural estimate from the Schwinger pair-production analogy gives:

$$v_{\text{perm}} \sim \frac{c}{\sqrt{\pi}} \approx 0.56 c \quad (15)$$

This corresponds to $r_{\text{eq}} \approx 1.26 \xi$ —a modest expansion of the core beyond the healing length. Quantifying v_{perm} from first principles is the key open calculation (Section 10).

7. Significance

7.1 Electron Mass from Medium Properties

The electron mass is determined by m_A , ρ_A , and c —all properties of the Aether medium—through a specific transcendental equation (Eq. 11) with no adjustable structure. The mass arises from three interlocking constraints:

- **Circulation quantization:** $\kappa_e = h/(2m_A)$ fixes the flow speed at each radius
- **Quantum minimum size:** $r_{\text{loop}} = \hbar/(m_e c)$ identifies the loop radius with the reduced Compton wavelength (Paper 7 [6])

- **Self-consistency:** m_e must equal the energy of a vortex ring of size r_{loop} , which itself depends on m_e

For $m_A \approx 628\text{--}689$ MeV (within the convergence zone), the corrected prediction matches the observed 0.511 MeV. The inputs m_A and ρ_A are independently constrained by QCD bridge physics [5] and the healing-length convergence [6]. The core energy fraction f_{core} is the remaining model-dependent quantity; computing the vortex core structure in the logarithmic potential would pin down both f_{core} and the best-fit m_A simultaneously.

7.2 Self-Regulating Mass

The bound-core model explains *why* the electron has the mass it does: the DCE mechanism creates mass until it shuts itself off. Every electron reaches the same equilibrium because every minimum-circulation vortex ring has the same κ_e , the same velocity profile, and therefore the same threshold crossing point. Mass universality follows from circulation quantization.

7.3 Open Tube / Closed Tube Taxonomy

The bound electron core ring is a *closed tube* of gently compressed Aether, stabilized by the encircling vortex circulation. The QCD bridge is an *open tube* of strongly compressed Aether, stabilized by quark vortex endpoints [5]. Both are topologically-stabilized bound Aether structures in the same medium, differing in topology (closed vs. open), compression ($\ll 1$ to ~ 5 vs. $\sim 7 \times 10^7$), and energy scale (0.5 MeV vs. ~ 1 GeV/fm).

7.4 What Is Derived vs. Assumed

The self-consistent mass equation (Eq. 11) is derived from the Donnelly vortex energy formula [7], circulation quantization ($\kappa_e = h/(2m_A)$), and the identification $r_{\text{loop}} = \hbar/(m_e c)$ from Paper 7 [6]. The Bernoulli correction (Eq. 8) is derived from the density-pressure coupling in the unbound superfluid. The formation narrative and energy conservation (Eq. 3) are derived from the DCE mechanism and Bernoulli pressure gradient.

Three quantities remain undetermined from first principles: m_A (constrained to 500–850 MeV/ c^2 by QCD bridge physics and healing-length convergence, but not uniquely determined), f_{core} (the core energy fraction, which requires solving the GPE with the logarithmic potential), and $r_{\text{loop}} = \hbar/(m_e c)$ (the loop radius, identified with the Compton wavelength in Paper 7 but not derived from vortex structure). Deriving r_{loop} from first principles remains an open problem (Section 10). Independently determining m_A or f_{core} would reduce the remaining freedom.

8. Relationship to Paper 7

This working paper proposes three modifications to the electron model in Paper 7:

1. **Core interpretation.** The vortex core is filled with bound Aether rather than being a depleted vacuum. The condensate density still drops to zero at $\sim \xi$; the core region contains non-condensate material (bound Aether) rather than nothing.
2. **Energy corrections.** The Donnelly formula undercounts the total energy by $\sim 5\text{--}14\%$ (core energy, model-dependent) and overcounts by $\sim 1.7\%$ (Bernoulli density deficit). The net correction shifts the best-fit m_A from ~ 600 MeV to $\sim 628\text{--}689$ MeV, still within the convergence zone.
3. **Mass origin.** The electron mass is not merely *determined* by the vortex equation—it is *selected* by the self-regulating DCE mechanism. The formation narrative provides a physical reason why the vortex settles at this particular energy.

No equations in Paper 7 need to be changed. The Donnelly formula remains the dominant contribution ($\sim 85\text{--}95\%$ of the total mass). The corrections refine the numerical prediction and shift the optimal m_A within the existing allowed range.

9. The Form Factor Challenge

Electron scattering experiments (Bhabha scattering at LEP, Møller scattering, $g-2$ measurements) find no deviation from the QED point-particle prediction up to momentum transfers $Q \sim 200$ GeV, corresponding to a spatial resolution of $\sim 10^{-18}$ m [9]. Yet the vortex model predicts an electron with loop radius $r_{\text{loop}} \approx 386$ fm $= 3.9 \times 10^{-13}$ m—five orders of magnitude larger. This section examines what the experiments actually constrain and how the vortex model may be consistent.

9.1 What the Experiments Measure

These experiments do not image the electron. They measure scattering cross-sections and compare them to QED predictions for a structureless point particle. “Point-like to 10^{-18} m” means: *the electron’s electromagnetic interactions match QED predictions at all tested momentum transfers*. Any internal structure would modify the electron–photon vertex, producing a form factor $F(Q^2) \neq 1$ that changes the cross-section at the scale $Q \sim \hbar c/r$, where r is the charge radius.

For comparison, the proton has a charge radius of 0.87 fm and its form factor deviates from 1 at $Q \sim 200$ MeV, exactly as expected from its spatial extent. The proton’s charge radius (0.87 fm) exceeds its Compton wavelength (0.21 fm) by a factor of 4, so structure is always resolvable.

For the electron vortex, if the charge were distributed around the 386 fm ring, $F(Q^2)$ would deviate from 1 at $Q \sim 0.5$ MeV. This is six orders of magnitude below LEP’s reach—the deviation would be enormous and unmissable. The absence of any deviation is a genuine constraint.

9.2 Topological vs. Geometric Charge

The resolution depends on the nature of the electron’s charge in the Aether model. Paper 4 [4] defines charge as “the direction of the Aether flow vector”—a “radial inflow of Aether toward the particle” that is “distinct from the circulatory vortex that produces the magnetic moment” and is “linear, non-rotational, and spherically symmetric around the charge.”

There are two possibilities:

1. **Geometric charge** (distributed on the ring). The radial inflow originates from the vortex tube distributed around the 386 fm circumference. At distances $\gg r_{\text{loop}}$, the flow averages to spherically symmetric, but at $r \lesssim r_{\text{loop}}$ the ring geometry is visible. This predicts $F(Q^2) \neq 1$ at $Q \sim 0.5$ MeV—ruled out by experiment.
2. **Topological charge** (from the winding number). The radial inflow is sourced by the vortex *topology*—the winding number itself, not the spatial geometry of the ring. The charge is a property of the topological defect, which has no spatial extent. This predicts $F(Q^2) = 1$ at all Q , consistent with experiment.

The distinction is analogous to a magnetic monopole: a monopole’s charge is topological (from the field configuration’s winding number), not geometric (from a spatial charge distribution). Its form factor is $F(Q^2) = 1$ because there is no charge distribution to resolve—the charge *is* the topology.

Paper 4’s description of charge as “spherically symmetric” radial inflow is consistent with option (2): the spherical symmetry is exact (not approximate), because it arises from topology rather than from geometric averaging of a ring source. The vortex ring carries mass (circulation KE) and spin (angular momentum) distributed over the 386 fm circumference, but the charge is a topological property of the vortex core with no spatial extent.

9.3 Status

Whether the electron’s charge is topological or geometric is the acknowledged open question from Paper 4 [4]: the microscopic mechanism connecting the vortex topology to the radial inflow has not been derived. If the charge is topological, the form factor challenge is resolved. If it is geometric, the vortex model is ruled out by existing scattering data. A first-principles derivation of the photon–vortex coupling in the condensate would settle this question.

10. Open Questions

1. **DCE permanence threshold.** What is v_{perm} ? A first-principles DCE calculation in vortex geometry would determine the equilibrium core size and the KE/ring mass partitioning. This is the key open calculation needed to make the model fully quantitative.
2. **Core energy in the logarithmic potential.** Computing the gradient and potential energy of a vortex core in the logarithmic condensate would determine f_{core} and, combined with the mass equation, pin down m_A uniquely.
3. **Muon and tau.** Could higher-energy vortex configurations (multiply-wound cores, knotted rings) stabilize at different masses through the same self-regulation mechanism? This is Paper 7 [6], Open Question 11.
4. **Form factor.** See Section 9 for a detailed analysis. The key open question is whether the electron's charge is topological (from the winding number) or geometric (distributed on the ring). If topological, $F(Q^2) = 1$ at all Q , consistent with experiment.
5. **Charge origin.** The bound core ring surface is a permanent DCE boundary with intense transient cycling. Could this be the microscopic mechanism that implements the radial Aether flow defining electric charge—an acknowledged gap in Paper 4 [4]?
6. **Cosmological formation.** What conditions in the early universe create vortex rings with the minimum circulation quantum? What determines the initial r_{loop} , and how does it relax to $\hbar/(m_e c)$?

References

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